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Problem Statement ID – 25013
Problem Statement Title – Real-Time Public Transport Tracking for Small Cities
Theme – Transportation & Logistics
PS Category – Software

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IDEA TITLE

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We propose developing a mobile and web-based platform that provides real-time bus tracking using GPS integration, enabling commuters to view live bus locations, estimated arrival times, and seat availability. Conductors can manage journeys, set routes, and update stops. Integration with maps APIs allows optimized route planning and improved travel time prediction.

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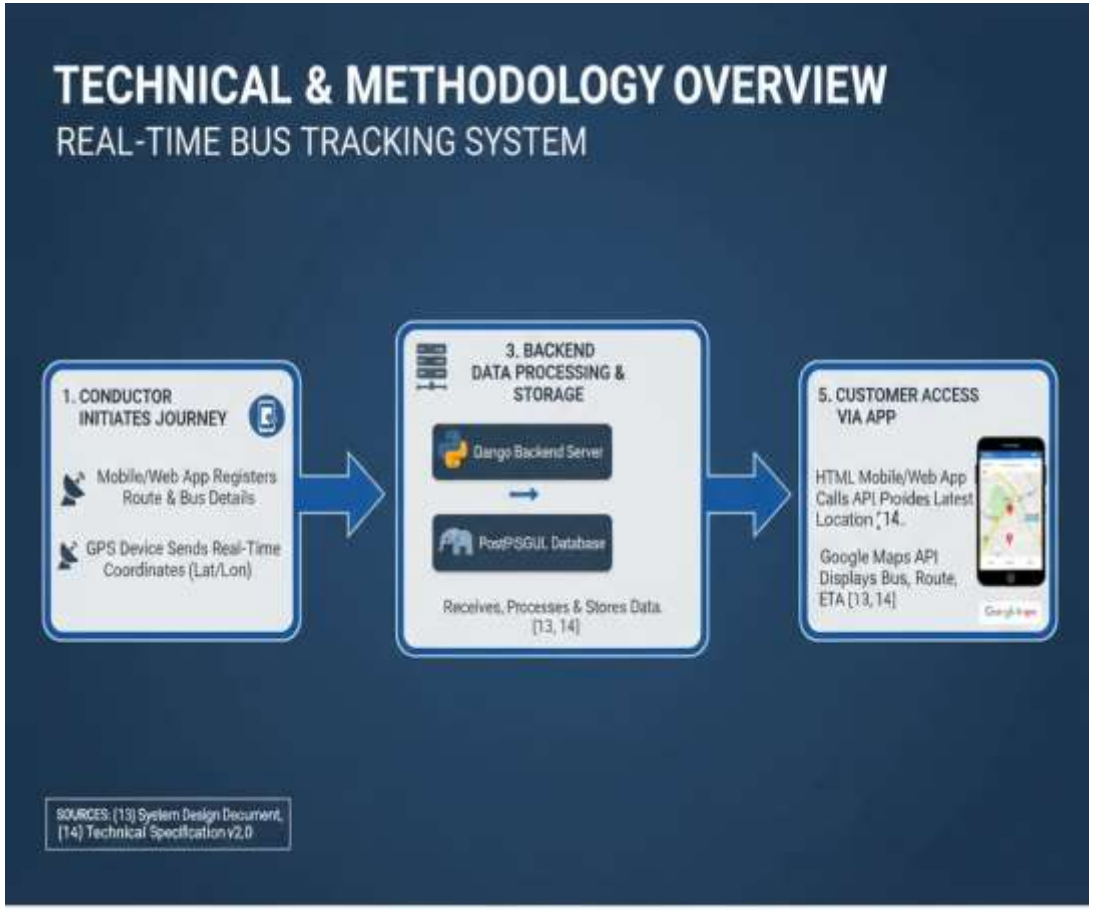
TECHNICAL APPROACH



Technologies: Django(Back-End),HTML(Front-End) PostgreSQL(Storage), Google Maps API, GPS tracking devices.

Methodology: Conductor registers journey → GPS-enabled buses transmit live location → Data stored in PostgreSQL → API serves real-time data → Customers access via mobile/web app.

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FEASIBILITY AND VIABILITY

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Feasibility: Requires low-cost GPS devices and internet-enabled buses.

Challenges: Network connectivity in rural areas, hardware costs, adoption resistance.

Mitigation: Use offline caching, government subsidies for GPS units, awareness campaigns.

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IMPACT AND BENEFITS

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Social: Improved commuter convenience and trust in public transport.
Economic: Increased public transport adoption reduces fuel expenses and congestion.
Environmental: Reduced traffic and pollution via lower private vehicle dependency.

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References: Case studies from state transport systems.
Research on Intelligent Transport Systems (ITS), Google Maps Transit API documentation.